



EXAM PAPERS PRACTICE

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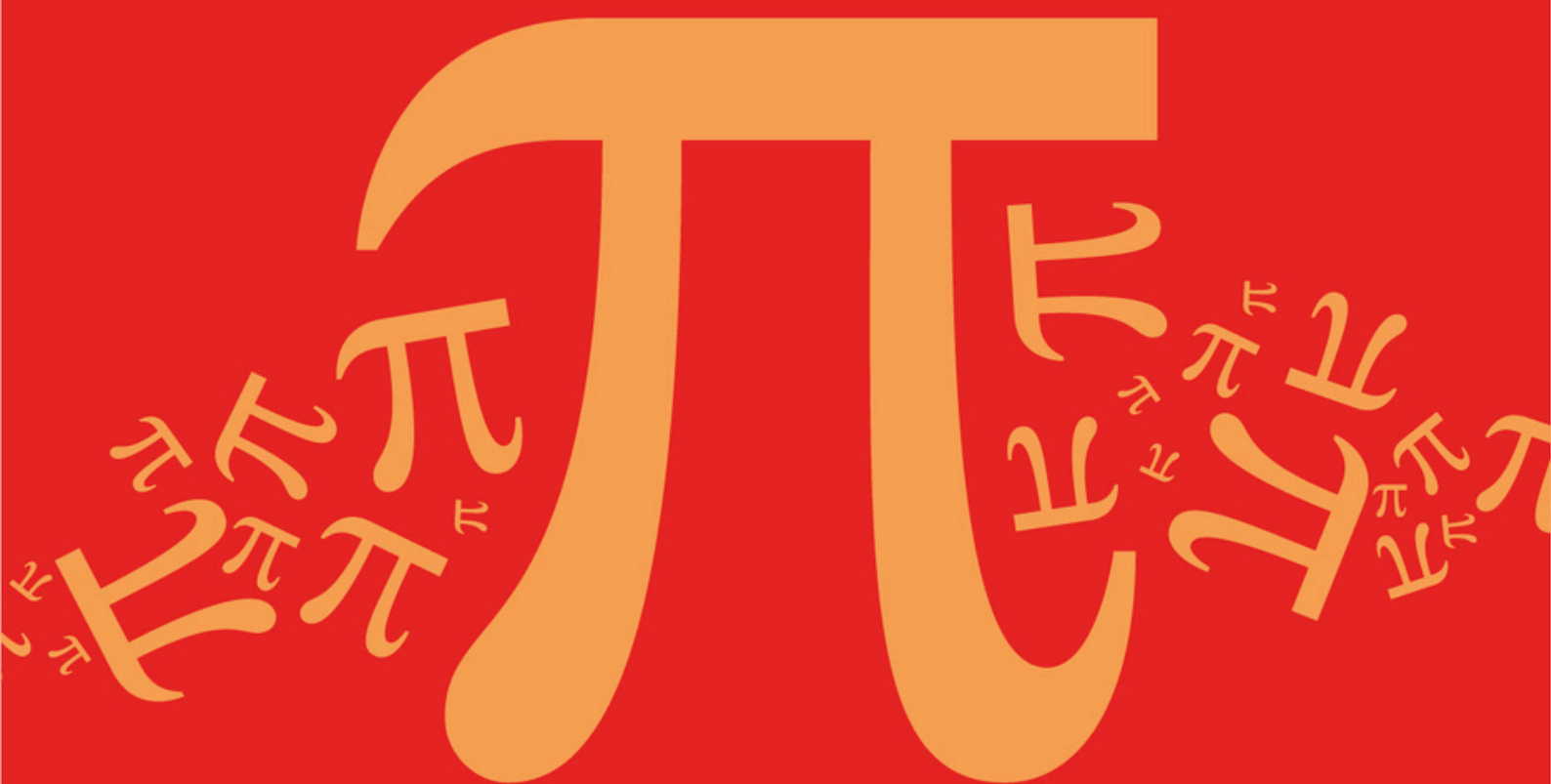
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C9: Factorise Determinants

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C9: Factorise Determinants

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

S is a singular matrix such that

$$\det S = \begin{vmatrix} a & a & x \\ x-b & a-b & x+1 \\ x^2 & a^2 & ax \end{vmatrix}$$

Express the possible values of x in terms of a and b .

(Total 7 marks)

Q2.

Let $\Delta = \begin{vmatrix} yz & xz & xy \\ x & y & z \\ x^2 & -y^2 & z^2 \end{vmatrix}$.

(a) Show that $(y + z)$ is a factor of Δ .

(2)

(b) Factorise Δ as completely as possible.

(4)

(Total 6 marks)

Q3.

Let $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ x & y & z \\ y+z & z+x & x+y \end{vmatrix}$.

(a) Use a row operation to show that $(x + y + z)$ is a factor of Δ .

(2)

(b) Hence, or otherwise, express Δ as a product of linear factors.

(2)

(Total 4 marks)

Q4.

Let $\Delta = \begin{vmatrix} n(n+1) & n+1 & -1 \\ 0 & 1 & n \\ 1 & -(n+1) & 1 \end{vmatrix}$.

(a) (i) Show that $(n^2 + n + 1)$ is a factor of Δ .

(2)

(ii) Hence, or otherwise, express Δ in factorised form.

(2)

- (b) By expanding Δ directly, show that

$$\Delta = [n(n+1)]^2 + f(n)$$

where $f(n)$ can be expressed as the sum of two squares.

(2)

- (c) Hence express the number 12 321 as the sum of three squares.

(2)

(Total 8 marks)

Q5.

(a) It is given that $\Delta = \begin{vmatrix} 16-q & 5 & 7 \\ -12 & -1-q & -7 \\ 6 & 6 & 10-q \end{vmatrix}$.

- (i) By using row operations on the first two rows of Δ , show that $(4-q)$ is a factor of Δ .

(2)

- (ii) Express Δ as the product of three linear factors.

(4)

(b) It is given that $\mathbf{M} = \begin{bmatrix} 16 & 5 & 7 \\ -12 & -1 & -7 \\ 6 & 6 & 10 \end{bmatrix}$.

- (i) Verify that $\begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix}$ is an eigenvector of $\mathbf{M}$ and state its corresponding eigenvalue.

(3)

- (ii) For each of the other two eigenvalues of $\mathbf{M}$, find a corresponding eigenvector.

(7)

- (c) The transformation T has matrix $\mathbf{M}$. Write down cartesian equations for any one of the invariant lines of T .

(2)

(Total 18 marks)

Q6.

Factorise fully the determinant $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & zx & xy \end{vmatrix}$.

(Total 8 marks)

Q7.

- (a) Expand the determinant

$$D = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ z & x & y \end{vmatrix}$$

(2)

- (b) Show that $(x + y + z)$ is a factor of the determinant

$$\Delta = \begin{vmatrix} x & y & z \\ y-z & z-x & x-y \\ x+z & y+x & z+y \end{vmatrix}$$

(2)

- (c) Show that $\Delta = k(x + y + z)D$ for some integer k .

(3)

(Total 7 marks)

Q8.

By considering the determinant

$$\begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix}$$

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show that $(x + y + z)$ is a factor of $x^3 + y^3 + z^3 - kxyz$ for some value of the constant k to be determined.

(Total 3 marks)

Q9.

- (a) Show that $(a - b)$ is a factor of the determinant

$$\Delta = \begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ bc & ca & ab \end{vmatrix}$$

(2)

- (b) Factorise Δ completely into linear factors.

(5)

(Total 7 marks)

Q10.

Factorise completely the determinant $\begin{vmatrix} y & x & x+y-1 \\ x & y & 1 \\ y+1 & x+1 & 2 \end{vmatrix}$.

(Total 6 marks)

Q11.

(a) Show that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)$$

(5)

(b) (i) Hence, or otherwise, show that the system of equations

$$\begin{aligned} x + y + z &= p \\ 3x + 3y + 5z &= q \\ 15x + 15y + 9z &= r \end{aligned}$$

has no unique solution whatever the values of p , q and r .

(2)

(ii) Verify that this system is consistent when $24p - 3q - r = 0$.

(2)

(iii) Find the solution of the system in the case where $p = 1$, $q = 8$ and $r = 0$.

(5)

(Total 14 marks)

Q12.

Express the determinant $\begin{bmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{bmatrix}$ as the product of four linear factors.

(Total 6 marks)

Q13.

(a) Show that $(a + b + c)$ is a factor of

$$\begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix}$$

Express this determinant as the product of $(a + b + c)$ and a quadratic factor.

(5)

- (b) Hence, or otherwise, show that there is a single real value of a for which the system of equations

$$ax + 3y + z = -5$$

$$4x + (1 + a)y + (a + 3)z = 9$$

$$2x + (1 - a)y + (a - 3)z = 15$$

does not have a unique solution, and find this value of a .

(5)

(Total 10 marks)



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